

Sultanate of Oman

Nama Water Services Company SAOC (NWS / OWWSC)

Consultancy Services for Design and Supervision for STP, Sewer & TE Networks Systems in Ibri

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Tutorial T01 — Sewage Flow and Pollution Load Calculation

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Renardet S.A. & Partners Consulting Engineers

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1. PURPOSE AND DESIGN BASIS

This tutorial sets out, step by step, how sewage flows and pollution loads are calculated for the Ibri project, from population through to the flows and loads that size the collection network and the sewage treatment plant. It is intended as a compact, self-contained method statement: every numerical value is taken from the NWS design guidelines or from the project Inception Report R0 demand model, and each is cited at the point of use.

1.1. REFERENCE DOCUMENTS AND CITATION CONVENTION

Abbreviation	Document
p##	PAM-GUD-203 — Wastewater Design Guidelines v1.0 (Rev 01), page ##
G1-p##	PAM-GUD-201 — General Design Guidelines v1.0 (Rev 01), page ##
G2-p##	PAM-GUD-202 — Potable Water & TSE Design Guidelines v1.0 (Rev 01), page ##
R0	Inception Report R0 demand workbook (Ibri Sewer Demand R0, August 2026)

Where the guidelines and the R0 model differ, both values are stated and the difference is flagged for confirmation with NWS (Section 4).



2. THE CALCULATION CHAIN

The calculation proceeds in nine steps. Steps 1-5 build the average flow; Step 6 converts it to peak flows for hydraulic sizing; Step 7 establishes the pollution loads independently of the flows; Steps 8-9 combine them at the treatment plant.

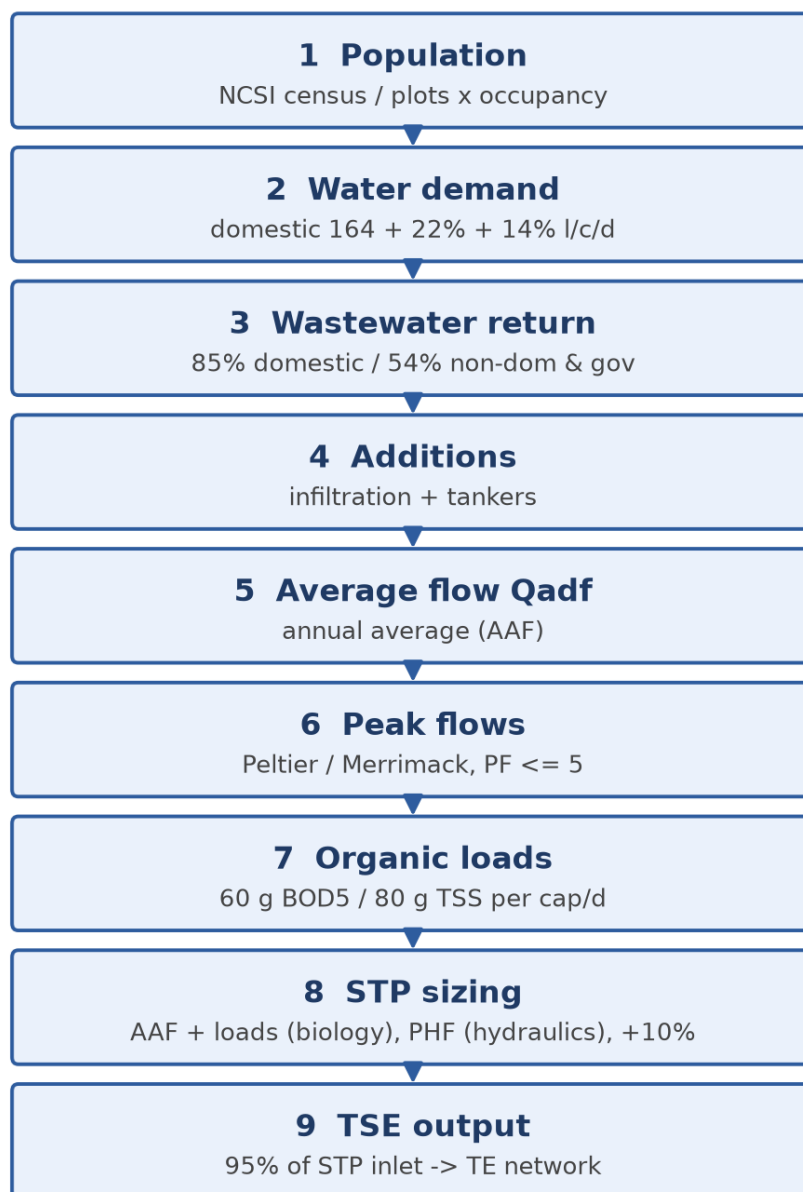


Figure 1 — Sewage flow and load calculation chain



2.1. STEP 1 — POPULATION

Two routes are used together (G1-p58). The census route projects the NCSI wilayat population and disaggregates it to settlements; it governs the dated model years (2030, 2055). The plot route multiplies the number of plots by the average properties per plot and the occupancy rate; it governs the ultimate (saturated) horizon, because saturation no longer depends on the growth rate.

Route	Formula / source	Governs
Census (top-down)	NCSI wilayat series, settlement disaggregation. Ibri: 183,564 (2024), growth 2.4-3.0 %/yr (R0)	Model years 2030 / 2055
Plots (bottom-up)	Population = plots x properties per plot x occupancy rate (G1-p58)	Ultimate / saturation

The scope horizons are the start year, 2030, 2055 and ultimate, at 5-year projection intervals; the design horizon is completion + 25 years or saturation, whichever governs (TOR p3, p14-15). The occupancy rate per settlement is to be confirmed from NCSI housing statistics.

2.2. STEP 2 — WATER DEMAND

Water demand is built from the domestic per-capita consumption, increased by fixed governorate ratios for non-domestic and governmental use (G1-p60-61). For Adh Dhahirah:

Component	Value (l/c/d)	Reference
Domestic	164 (R0 computes 163.5 from 2021-24 billing — equivalent)	G1-p60, R0
Non-domestic (+22% of domestic)	36.1	G1-p60-61
Governmental (+14% of domestic)	23.0	G1-p60
Total water demand	223	—

For large individual facilities (schools, hospitals, malls) the unit rates of G1-p61 Table 12 may replace the percentage approach (e.g. 130 l/pupil/d for schools, 650 l/bed/d for hospitals).

2.3. STEP 3 — WASTEWATER RETURN

Only part of the supplied water returns to the sewer. The return rates of G1-p71 (Table 19) are 85% for domestic and tankered consumption and 54% for non-domestic and governmental consumption:



Stream	Return	Per-capita wastewater (l/c/d)
Domestic	85%	$164 \times 0.85 = 139.4$
Non-domestic + governmental	54%	$(36.1 + 23.0) \times 0.54 = 31.9$
Total wastewater generation	—	≈ 171

2.4. STEP 4 — ADDITIONS: INFILTRATION AND TANKERS

- Infiltration, new networks: 720 L/d per km of sewer (G1-p72). For existing inland networks: 10% of the wastewater flow; up to 40% for coastal/high-groundwater systems. No stormwater allowance — the system is strictly separate.
- Yellow tankers: currently about 17% of STP inflow nationally; design coverage reaches 100% by the end of the planning period (G1-p73). The R0 model collects tankered flows from settlements within 25 km of the STP.

Note: the R0 model currently applies 10% infiltration to selected settlements, whereas the guideline value for new networks is the linear 720 L/d/km — roughly an order of magnitude smaller for this network. Both are carried until NWS confirms the basis (Section 4).

2.5. STEP 5 — AVERAGE FLOW

The average daily flow (Qadf, equal to the annual average flow AAF of GUD-203) is:

$$\text{Qadf [m}^3\text{/d]} = \text{Population} \times 171 \text{ l/c/d} / 1000 + \text{Infiltration} + \text{Tanker deliveries}$$

GUD-203 (p65-66) distinguishes AAF (annual average), MDF (maximum day) and PHF (peak hour); each has a distinct use in Step 8.

2.6. STEP 6 — PEAK FLOWS

Peaking applies to the sewage component only; infiltration is not peaked. Two formulas are permitted (G1-p71-72):

Method	Expression	Result
Peltier (IMP 2024)	$PF = 1.5 + 1/\sqrt{Q_m}$, Q_m in L/s	Peak factor on average flow
Merrimack	$Q_{pdf} = 2.65 \times Q_{adf}^{0.879}$ (both ML/d, >100 properties)	Peak daily flow directly
Cap	Hourly PF ≤ 5.0	—



The two methods do not agree (the worked example gives PF 1.72 vs 2.48). Peltier is the current IMP 2024 method; the binding choice is to be confirmed with NWS. The R0 model additionally applies a +20% weekly peak, which has no guideline source and is likewise flagged.

2.7. STEP 7 — ORGANIC LOADS

Per-capita loads are fixed minimums (p74) and do not scale with water consumption:

Parameter	Value	Reference
BOD5	$\geq 60 \text{ g/cap/d}$	p74
TSS	80 g/cap/d	p74
COD (domestic)	$1.8 - 2.2 \times \text{BOD}$	p74

Concentration is always derived, never assumed: $C [\text{mg/l}] = \text{Load} [\text{kg/d}] \times 1000 / Q [\text{m}^3/\text{d}]$. Low per-capita water use produces concentrated sewage: expect BOD around 300-400 mg/l here. A result near 200 mg/l (European dilution) indicates an error.

2.8. STEP 8 — WHAT EACH FLOW IS USED FOR

Use	Governing flow	Additional rules
Gravity pipe sizing	Peak (hourly) flow	$d/D \leq 0.65$ ($D \leq 350$) / 0.50 ($D > 350$); $v = 0.75\text{-}3.0 \text{ m/s}$; Table 11 minimum gradients (p26-29)
Pumping stations / force mains	Peak flow with any one pump out	$v = 1.0$ (intermittent) - 2.5 m/s (p39, p50)
STP hydraulic pass-through	PHF	p65-66
STP biology	AAF + organic loads	p65-66
TE / TSE pipelines	TSE = 95% of STP inlet	Roughness penalty for TE: $\varepsilon +30\%$, C -10% (G2-p104)

2.9. STEP 9 — STP INCOMING FLOW AND OUTPUTS

STP incoming flow = network Qadf (including infiltration) + tankered flows + 10% operational allowance (p65 Table 29, G1-p73). A plant at or above 20,000 m³/d falls in the Large category (p65); per the TOR, the Phase I capacity relative to this threshold determines whether the new STP preliminary design remains in the consultancy scope.

- TSE production = 95% of STP inlet (G1-p73) — this feeds the TE network design.
- Sludge $\approx 0.25 \text{ kg}$ per m³ treated (R0 planning rate; to be refined at process design).



3. WORKED EXAMPLE

A settlement of 10,000 persons served by 25 km of new sewers, all values rounded:

#	Step	Calculation	Result
1	Population	given	10,000 cap
2	Water demand	164×1.36	223 l/c/d $\rightarrow$ 2,230 m ³ /d
3	Wastewater return	$164 \times 0.85 + 59.1 \times 0.54$	171.3 l/c/d $\rightarrow$ 1,713 m ³ /d
4	Infiltration (new network)	720 L/d/km $\times$ 25 km	+18 m ³ /d ($\approx$ 1%)
5	Average flow Q _{adf}	$1,713 + 18$	1,731 m ³ /d = 20.0 L/s
6a	Peltier peak	$PF = 1.5 + 1/\sqrt{20.0} = 1.72$	peak 34.5 L/s
6b	Merrimack peak	$2.65 \times 1.731^{0.879}$	4.29 Ml/d = 49.7 L/s (PF 2.48)
7	Loads BOD / TSS / COD	$10,000 \times 60 / 80 / 120$ g/d	600 / 800 / 1,200 kg/d
7b	Concentrations	load / Q _{adf}	BOD 347, TSS 462, COD 693 mg/l
8	STP incoming flow	$1,731 \times 1.10$	1,904 m ³ /d
9	TSE production	$1,904 \times 0.95$	1,809 m ³ /d

Table — Worked example, 10,000-person settlement

Sanity checks that should always hold: BOD concentration between 300 and 400 mg/l; peak factor between 1.5 and 5.0; infiltration far smaller than sewage flow for a new inland network.



4. GUIDELINE VS INCEPTION R0 — RECONCILIATION REGISTER

The R0 demand model and the guidelines agree on the core parameters; the differences below are carried openly until confirmed with NWS at the kick-off stage:

Item	Guideline (binding)	R0 adopted	Status
Domestic per-capita demand	164 l/c/d (G1-p60)	163.5 computed from billing	Consistent
Return ratios	85% / 54% (G1-p71)	Same	Consistent
Infiltration	720 L/d/km, new networks (G1-p72)	10% of wastewater flow	To reconcile
Peaking	Peltier / Merrimack, PF ≤ 5 (G1-p71-72)	+20% weekly peak added	Confirm basis
Tanker catchment	Not specified	25 km radius	NWS to confirm
STP margin / TSE ratio	+10% / 95% (G1-p73)	Same	Consistent
Sludge rate	Not specified	0.25 kg/m ³	R0 planning value

4.1. OPEN INPUTS

- Occupancy rate per settlement — NCSI housing statistics (partly available in the R0 workbook; value per settlement to be confirmed).
- Existing Ibri STP capacity, headworks invert and spare capacity.
- Model start year and confirmation of the 2030 / 2055 / ultimate horizons.

